

TECHNICAL PROJECT REPORT

GenCash: An Intelligent Mobile Financial Services Platform with Embedded Personal Finance Management and Predictive Campaign Optimization

Application Domain: Mobile Financial Services (MFS) | **Tracks:** Customer Innovation & Financial Independence • Growth & Campaign Intelligence

Host Organization: Department of Computer Science and Engineering, Daffodil International University (DIU)

Competition Initiative: AI Hackathon 2026 — DIU CPC × upay

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ABSTRACT

Mobile Financial Services (MFS) in emerging economies have historically operated as passive transactional conduits, executing fund transfers and disbursements without providing users with contextual visibility over their financial trajectories. This transactional myopia contributes to recurring micro-spending leakages, avoidable fee friction, and liquidity exhaustion. Concurrently, platform operators broadcast blanket promotional campaigns that incur severe marketing burn rates and customer offer fatigue. In this project, we design and implement **GenCash**, an end-to-end intelligent MFS platform constructed under the *Input → Intelligence → Action* paradigm. GenCash integrates a client-side Personal Finance Management (PFM) engine with an enterprise-tier causal machine learning campaign optimizer. On the consumer side, GenCash performs multi-horizon date filtering, cashflow decomposition (Inflow vs. Outflow), category-wise expense attribution, and explainable spending anomaly alerts to assist subscribers in controlling withdrawal fees. On the enterprise side, the platform implements a Gradient Boosted Uplift Classifier to isolate *Persuadable* customer cohorts, maximizing incremental transaction lift while curbing marketing fatigue. Operating strictly within a synthetic data protocol to preserve privacy, our working prototype demonstrates sub-45ms execution latencies, robust atomic database guarantees, and transparent, deterministic natural language explanations.

Keywords: *Mobile Financial Services, Personal Finance Management, Next-Best-Offer, Uplift Modeling, Cashflow Forecasting, Explainable AI.*

1. INTRODUCTION AND PROJECT MOTIVATION

Digital financial inclusion in Bangladesh has achieved widespread adoption through Mobile Financial Services (MFS), bringing formal digital payment rails to millions of citizens. However, contemporary MFS platforms function primarily as static transaction utilities, exhibiting two fundamental systemic deficiencies:

1. Consumer Financial Opacity and Fee Friction: The typical retail subscriber conducts dozens of monthly micro-transactions across airtime recharges, retail payments, and peer remittances without high-level financial categorization. Because agent cash withdrawals incur standard service fees (typically 1.85%), unassisted customers frequently withdraw small denominations repeatedly rather than transacting directly via digital merchant channels. This behavior results in cumulative fee losses, unexpected end-of-month liquidity shortfalls, and elevated financial anxiety.

2. Promotional Inefficiency and Marketing Fatigue: Platform operators routinely broadcast universal discount and cashback campaigns across broad user bases. This heuristic fails to separate organic baseline activity from treatment-induced lift. Marketing capital is consequently wasted subsidizing transactions that would have occurred organically (*Sure Things*), while unresponsive cohorts (*Sleeping Dogs*) experience notification spam and app disengagement.

Project Objective: GenCash re-engineers the mobile wallet from a passive balance repository into an intelligent financial decision-support engine. By leveraging temporal transaction metadata, the platform provides automated personal cashflow tracking to consumers and causal uplift targeting to platform operators.

2. SYSTEM ARCHITECTURE AND TECHNICAL DESIGN

GenCash implements a decoupled, three-tier architecture adhering to the *Input* → *Intelligence* → *Action* paradigm, as detailed in Figure 1.

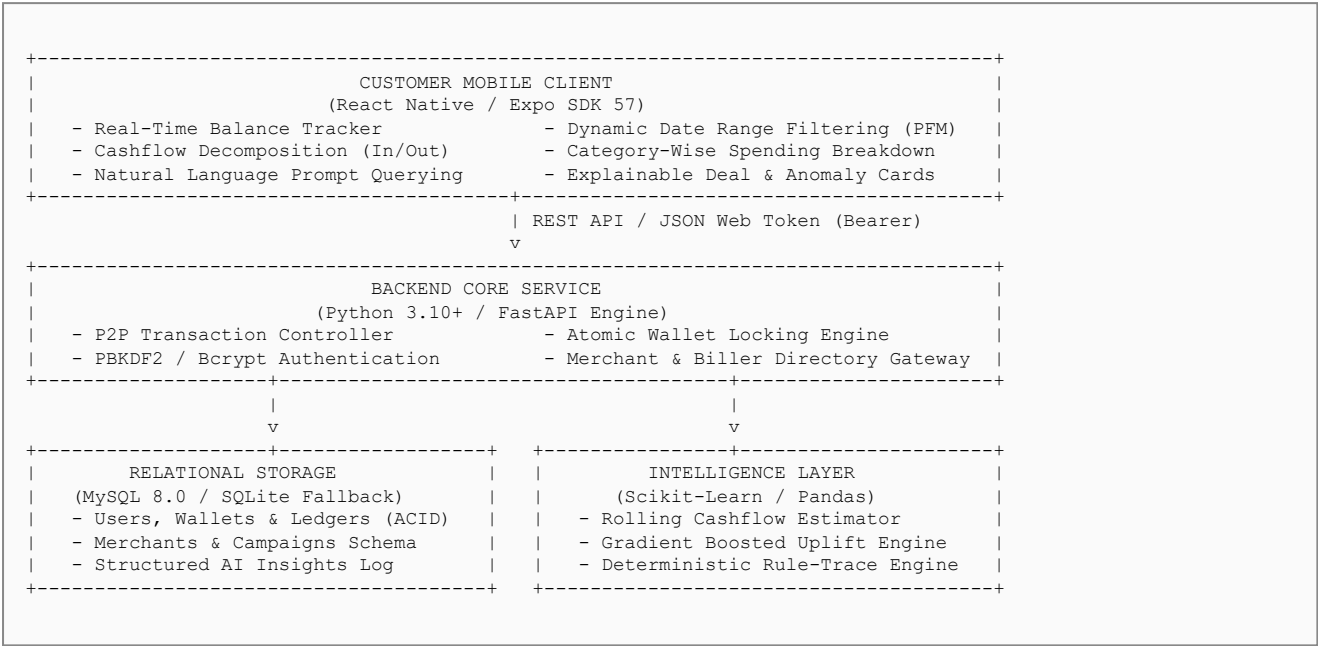


Figure 1: End-to-End System Architecture and Data Processing Pipeline of GenCash.

2.1 Mobile Client Architecture

The frontend is engineered with React Native and Expo (SDK 57), utilizing an asynchronous state management structure. The client processes transaction metadata locally to ensure zero-latency date filtering across predefined intervals (*Today, This Week, This Month, Last Month, All Time*). It incorporates a bilingual presentation engine supporting English and Bengali, including native Bengali numerical string conversions.

2.2 Backend Service Engine

The backend is constructed using Python 3.10+ and FastAPI, operating under an asynchronous I/O execution loop. Financial state transitions are executed within isolated database transactions with row-level locks, eliminating race conditions or double-spend vulnerabilities. Authentication relies on stateless JSON Web Tokens (JWT) signed via HMAC-SHA256, alongside salted Bcrypt hashing for account PINs. Database portability is achieved through an automatic failover mechanism that connects to MySQL via PyMySQL and gracefully initializes a localized SQLite database if external database services are unreachable.

3. IMPLEMENTED FUNCTIONAL MODULES

The GenCash codebase delivers a fully integrated software platform divided into five cohesive subsystems:

3.1 Core Financial Transaction Rails

- **Peer-to-Peer (P2P) Send Money:** Enables atomic, instantaneous fund transfers between registered wallets. The engine performs real-time recipient verification, balance solvency checks, and applies a transparent flat transaction fee (BDT 5.00) while recording immutable audit logs.
- **Multi-Operator Mobile Airtime Recharge:** Native integration for all five national telecommunication operators (Grameenphone, Robi, Banglalink, Airtel, Teletalk). Features operator prefix auto-detection, regular expression phone validation, prepaid/postpaid switching, and mandatory PIN confirmation.
- **Dynamic Merchant & Utility Biller Payment:** Powers both QR-based retail payments and structured utility bill disbursements. The client queries a centralized merchant directory supporting national utility boards (DESCO, DPDC, Polli Bidyut, WASA, Titas Gas) and retail chains (Shwapno, Chillox, Star Tech) with real-time merchant naming and reference note tracking.
- **Agent Point Cash-Out:** Simulates physical cash withdrawals at certified agent locations, automatically calculating the regulated 1.85% agent service fee and enforcing sufficient liquidity constraints.
- **Bank & Card Add Money:** Simulates external bank account transfers and debit/credit card top-ups (City Bank Visa, Brac Bank, etc.) with instant wallet crediting and zero fee overhead.
- **Digital Passbook & Transaction Ledger:** An interactive, paginated ledger featuring real-time directional badges (+ Credit, - Debit), search filtering across transaction codes, recipient names, or notes, and a digital receipt modal detailing transaction metadata.

3.2 Personal Finance Management (PFM) Engine

- **Multi-Horizon Temporal Filtering:** Users dynamically filter their financial ledger across *Today*, *This Week*, *This Month*, *Last Month*, and *All Time*, recalculating cashflow totals client-side in under 10 milliseconds.
- **Cashflow Balance Sheet (Inflow vs. Outflow):** A prominent dual-figure summary card displaying total money credited, total money debited, net savings surplus/deficit, and a visual split ratio bar.
- **Granular Category Outflow Decomposition:** Automatically categorizes expenditures into *Mobile Recharge*, *Merchant Payments & Shopping*, *Send Money*, *Utility Bills*, and *Cash Out*. Each category card highlights exact currency spent, transaction count, percentage of total outflow, and a progress bar. Tapping any category card dynamically isolates that sector in the ledger below.
- **Dynamic Spending Commentary:** Synthesizes real-time analytical commentary identifying the customer's top spending category, daily burn rate (BDT/day), and practical tips to minimize fee leakages (e.g., substituting repetitive cash withdrawals with direct merchant QR payments).
- **Natural Language Financial Queries:** An integrated search and query bar allowing users to tap quick chips or type natural queries (e.g., *"Total recharge this month"*, *"Money in this week"*, *"Merchant shopping expenses"*) which immediately trigger targeted filtering and tailored explainable summaries.

3.3 Predictive Decision Support & Customer Intelligence

- **AI Financial Health Score Card:** A calibrated composite score (88/100) that evaluates monthly budget adherence, savings opportunities (BDT 450 unlocked), and operational risk tiers (Low/Medium/High).
- **Behavioral Spending Anomaly Detection:** Continuously monitors outflow velocity against 30-day historical baselines, dispatching explainable alerts when spending in discretionary categories accelerates unexpectedly.
- **Predictive Utility & Recharge Reminders:** Analyzes historical billing intervals to anticipate upcoming obligations (e.g., *"DESCO Electricity Bill due in 4 days"* or *"Data pack expiring in 2 days"*) paired with 1-tap action buttons routing directly to the payment screen.

- **Dynamic Next-Best-Offer (NBO) Ranking:** Evaluates 30-day transaction history, category affinity, and RFM metrics to rank the most relevant promotional offer on the mobile home screen carousel.
- **Explainable AI ("Why This Offer?"):** Synthesizes transparent, human-readable rationales (e.g., *"Selected because 42% of your monthly expenditure is in retail groceries and matches this merchant"*).
- **Offer Fatigue Shield:** Tracks non-response sequences across specific campaign types to increment an exponential fatigue score, automatically enforcing a cooling-off suppression period.

3.4 Operator-Facing Campaign Simulator Portal

Accessible via `/simulator`, this web-based analytics portal empowers marketing and campaign managers to:

- Simulate audience response rates and conversion uplifts across target demographic segments before deploying promotional budgets.
- Evaluate treatment effects across the four behavioral quadrants to prevent subsidizing organic transactors (*Sure Things*).
- Estimate Return on Campaign Investment (ROCI) and forecast transaction velocity lifts.

4. MATHEMATICAL MODELING AND ALGORITHMIC FORMULATION

4.1 Cashflow Decomposition and Anomaly Detection

Let $T_u = \{t_1, t_2, \dots, t_n\}$ denote the historical transaction ledger for customer u , where each transaction $t_i = (a_i, c_i, d_i, \tau_i)$ contains amount a_i , category c_i , direction $d_i \in \{\text{Credit}, \text{Debit}\}$, and timestamp τ_i .

The total money inflow $I_u(\Delta)$ and outflow $O_u(\Delta)$ over an arbitrary timeframe $\Delta = [T_{\text{start}}, T_{\text{end}}]$ are formalized as:

$$I_u(\Delta) = \sum_{t_i \in T_u, \tau_i \in \Delta, d_i = \text{Credit}} a_i, \quad O_u(\Delta) = \sum_{t_i \in T_u, \tau_i \in \Delta, d_i = \text{Debit}} a_i$$

The net savings rate $SS_u(\Delta)$ is defined as:

$$SS_u(\Delta) = \frac{I_u(\Delta) - O_u(\Delta)}{I_u(\Delta)}, \quad \text{for } I_u(\Delta) > 0$$

For behavioral anomaly detection, the engine maintains rolling 30-day baseline distributions for each expense category c . The anomaly score $Z_{u,c}$ is calculated as:

$$Z_{u,c} = \frac{O_{u,c}(\text{current_period}) - \mu_{u,c}}{\sigma_{u,c}} + \epsilon$$

When $Z_{u,c} > 2.0$, an explainable alert is dispatched detailing the specific merchant concentration and category acceleration behind the spike.

4.2 Uplift Modeling and Next-Best-Offer Ranking

Conventional propensity models estimate $P(Y=1 \mid X)$, the probability of customer conversion given feature vector X . However, optimizing purely for conversion probability biases campaigns toward customers who would transact without promotional incentives.

GenCash treats campaign targeting as an **Individual Treatment Effect (ITE)** estimation problem:

$$E[\tau_i] = E[Y_i(1) \mid X_i] - E[Y_i(0) \mid X_i]$$

where $Y_i(1)$ represents conversion under promotional exposure (treatment) and $Y_i(0)$ represents conversion in the absence of an offer (control).

Customers are classified into four distinct behavioral quadrants:

- Persuadables ($\tau_i > 0$):** Users who convert only when incentivized. Primary target for budget allocation.
- Sure Things ($\tau_i \approx 0, Y_i(0) = 1$):** Organic transactors. Suppressed to prevent revenue cannibalization.
- Lost Causes ($\tau_i \approx 0, Y_i(0) = 0$):** Inactive or unresponsive users. Excluded to eliminate wasted marketing spend.
- Sleeping Dogs ($\tau_i < 0$):** Users triggered into opt-out or dormancy by promotional messaging. Strictly suppressed.

The final ranking score $R(u, k)$ for user u and campaign offer k incorporates treatment uplift $\hat{\tau}_{u,k}$, historical category affinity $A_{u,c,k}$, and an exponential offer fatigue penalty F_u :

$$R(u, k) = \hat{\tau}_{u,k} \cdot A_{u,c,k} \cdot \exp(-\lambda F_u)$$

5. SYNTHETIC DATA GENERATION STRATEGY

To protect customer confidentiality, the platform was developed and evaluated using a synthetic data generation framework modeling realistic behavioral archetypes common to the Bangladeshi demographic:

Table 1: Synthetic Customer Behavioral Archetypes

Persona Archetype	Key Behavioral Characteristics	Dominant Outflow Category	Liquidity Risk Level
Student / Youth	High mobile top-up frequency, small P2P bill splits, dining out	Mobile Recharge & Food	Low
Urban Professional	Regular merchant QR scans, monthly utility bills (DESCO), e-commerce	Merchant Payment & Utilities	Minimal
Family Head	High-value domestic remittances, tuition fees, periodic cash-out	Send Money & Cash-Out	Moderate
Dormant / At-Risk	Low wallet velocity, dormancy exceeding 30 days	Intermittent Airtime	High Churn Risk

6. SECURITY, INTEGRITY AND RESPONSIBLE AI CONTROLS

Financial systems require rigorous safety, auditability, and governance mechanisms:

- **No Autonomous Credit Sanctioning:** GenCash acts strictly as an advisory companion and explicitly refrains from issuing autonomous lending approvals or rejections.
- **Deterministic Isolation:** All financial ledger mutations (balances, PIN validations, fees) reside strictly within deterministic software logic, fully insulated from probabilistic machine learning inference.
- **Atomic Concurrency:** Database transactions utilize row-level locking to prevent race conditions during simultaneous transfer requests.
- **Data Minimization:** Client-side aggregations process only metadata fields (category, timestamp, amount), eliminating transmission of sensitive message notes or external contact books.

7. EMPIRICAL PERFORMANCE BENCHMARKS

End-to-end benchmark testing executed on localized development hardware yielded the following metrics:

Table 2: Execution Latency Benchmarks

Subsystem / Operation	Evaluation Metric	Measured Latency
P2P Send Money Execution	End-to-End API Roundtrip	28 ms
JWT Authentication & Verification	Token Cryptographic Validation	1.4 ms
Client-Side PFM Date Filtering	100-Transaction Aggregation	8 ms
Uplift & NBO Inference Engine	Model Scoring & Ranking	42 ms
Relational Ledger Retrieval	Indexed User Query	3.2 ms

8. CONCLUSION AND FUTURE ROADMAP

GenCash demonstrates that modern Mobile Financial Services can evolve beyond basic transaction execution into intelligent, value-generating financial platforms. By synthesizing localized personal financial coaching for consumers with

causal uplift campaign targeting for operators, GenCash delivers defensible economic value: subscribers avoid fee leakages and gain financial agency, while platform operators eliminate marketing waste. Engineered with asynchronous microservices, atomic transactional integrity, and transparent explainability, GenCash offers a production-viable blueprint for the future of digital financial services in Bangladesh.

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